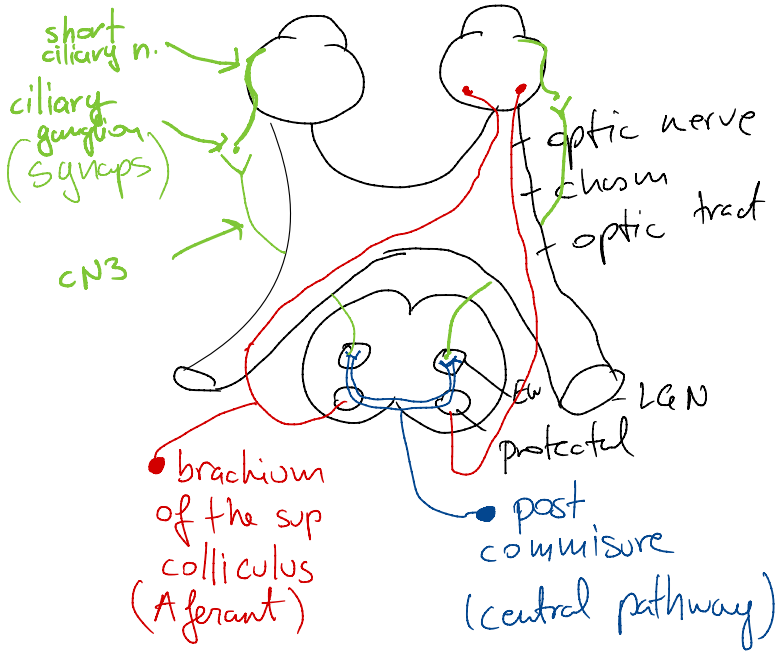
Pupillary Pathway

Figures used: 14.11, 14.12, 14.13, 14.3, 14.16, 14.19, 14.17, 14.18, 14.21

afferent – sensory (take info to)
efferent – motor (info away from)

I. Pupillary pathway

A. Afferent pathway follows the visual fibers to the optic tract



1. The pupillary fibers leave the optic tract and travel within the brachium of the superior colliculus to the pretectal area of the midbrain.

B. Central pathway: fibers travel from the pretectal nucleus to the Edinger Westfall preganglionic cells

C. Efferent pathway: follows cranial nerve 3

D. Clinical correlate: Direct and consensual pupillary light response

1. Direct pupillary light response: when we shy in the eye and looking the same eye. constriction of ipsilat. eye
2. Consensual pupillary light response: the other eye that is not directly "looked" constricton of contract eye

II. Disruption of the afferent pathway

A. Disruption of the afferent pathway will affect the direct response of the ipsilateral eye and the consensual response of the contralateral eye.

B. Swinging flashlight test

1. The patient fixates on a distance target, and the doctor moves a light from one eye to the other several times.
2. It is important to make sure each pupil is illuminated with an equal amount of light.
3. If there is no damage to the visual pathway, there will be little or no change in pupil size when switching the light from one eye to the other.
4. If there is damage to the afferent pathway, both pupils will dilate quicker when shining light into the pupil on the side ipsilateral to the damage.

Hypos – should be the same in the bouth eyes.
(AI CHECK) (Constriction of the pupil.)

C. Cause of RAPD: RELATIVTLY AFFERENTNI PUPIL DEFECT
the one dialeted quicker is a defect

1. Retina can couse it, but it have to be quit large
(se gnifficant amount of retina)

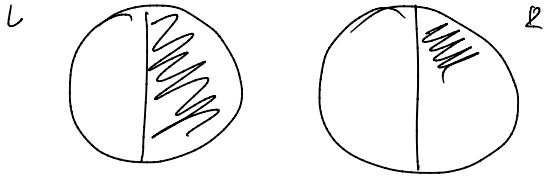
2. Optic nerve small lesion will couse a large RAPD
↳ will look really normal



AI CHECK

3. Optic tract (homonomosia robi)
visual defect in both eyes on the same side

Left RAPD (there more fibers defected)



4. Superior brachium
contra lateral RAPD
- normal vision (vision field) : monosklerosa
you will still have a diffect

5. Clinical correlate: media opacities do not cause an ipsilateral RAPD
important to know!
So we do not want to say cataract is why we don't see

III. Disruption in the central nervous system pathway

A. Lesions of the midbrain can affect the fibers between the pretectal and Edinger-Westphal preganglionic cells

B. Light-near dissociation light response it's absent

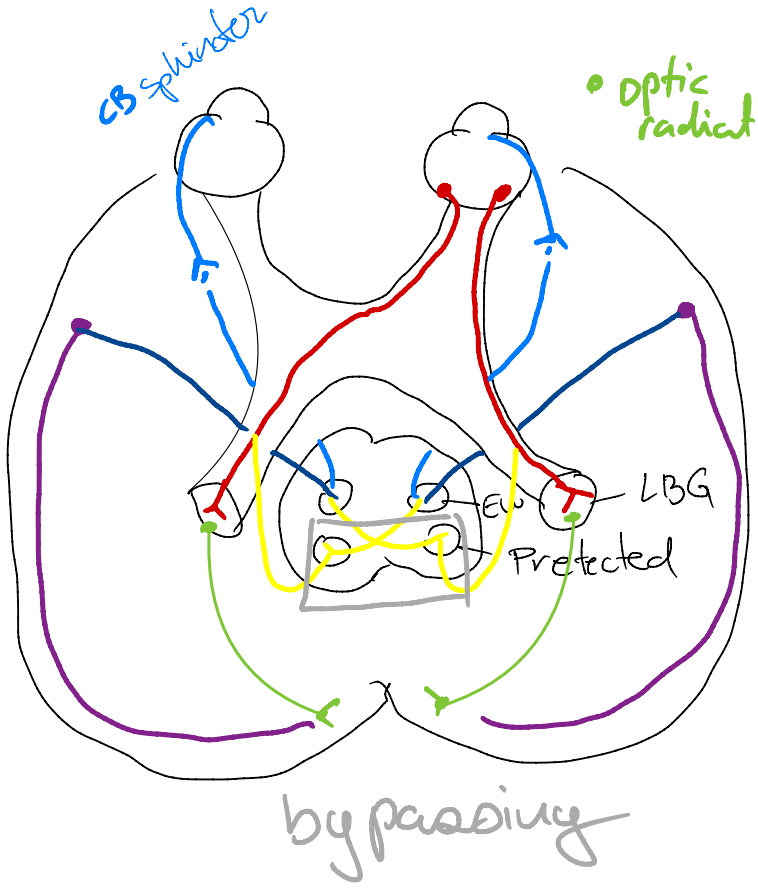
1. Near pupil response bypasses the central portion of the pupillary light response

2. Near pupil response pathway

a. Afferent

b. CNS

c. Efferent



signal frontal field (central path.)

will effect forsight but nearsight & accomodation not because it goes completly different way.

3. Clinical correlate: Argyll Robertson pupil

associated with neurosyphilis

a. Bilateral light near dissociation

b. Without any light stimulus, the pupils will be very small. !?

c. Most commonly due to neurosyphilis

4. Clinical correlate: dorsal midbrain syndrome

back of the midbrain

a. Most commonly due to a pineal gland abnormality

b. Bilateral light near dissociation

c. Pupils are normal to mid-dilated !?

IV. Disruption in the efferent parasympathetic pathway

A. A parasympathetic lesion will cause the ipsilateral eye to show poor direct, consensual, and near pupillary responses.

problem with sphincter — nothing gets the pupile "move" also no accomodation

1. Ipsilateral mydriasis — miosis

2. Anisocoria worse in bright light
   ↳ different size of pupils

B. Causes

1. Cranial nerve 3 palsy
   ptosies
   SR, MR, IR, IO, levator

2. Tonic pupil — problem with post ganglion fibers

   a. Caused by damage to the ciliary ganglion or short ciliary nerves

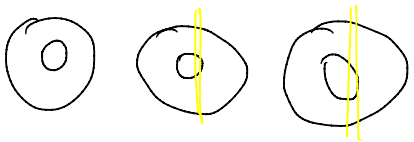
      (1) Autonomic dysfunction
      diabetis - systemic cause

      (2) Orbital mass, trauma, or surgery
      tumor in orbital, ↳ retinal detachment surgery one of the most common reasons

      (3) Idiopathic (Adie tonic pupil)
      when we do not know what caused it. Irregular shaped pupil.

Parasympathetic will cause problems with accommodation

   b. Segmental constriction.
      video where pupil wheant on 5 o'clock when lightened.



c. Decreased corneal sensitivity

often times will get

d. Cholinergic denervation supersensitivity

with tonic pupil you can put 0.12% pilocarpi
never gonna construct normal pupile
but will supersensitivity

3. Pharmacologically dilated pupil

a. Pharmacologic testing: 1% pilocarpine

↑ constriction

Doesn't affect the muscles or eyelids.

↳ constrict any pupile except when pharmacological dilated pupil

↳ we are directly stimulating it with pilocarpine

scapolamine patch - treat disiness on the boats or on the field because of chemicals

V. Disruption of the efferent sympathetic pathway: Horner syndrome

A. Miosis

B. Anisocoria greater in dim light

C. Normal reaction to light
also the flash light test

D. Dilation lag
Normaly we don't test it.
Usually you do it after swinging flash light and thake the light out of the eye.
more anisocoria at 5 sec vs. 15 sec

E. Ptosis
Because of Muller muscle, not levator (swoth m.)
(look for light reflection vs. eye lid)

F. Facial anhidrosis (absence of sweating)
Usually not gonna tell you that they are sweating because you don't notice it.

G. Pathway

1. Central neuron (first order neuron)

   a. Hypothalamus → brainstem → cervical spinal cord

cause of Horner's syndrom: stroke, multiple sclerosa (MS)

2. Preganglionic neuron (second order neuron)

   a. Cervical spinal cord → apex of the lung → superior cervical ganglion

cause of Horner's syndrom: thoracic injury (surg

Pancoast tumor (starting in liver, metastazijski)

3. Postganglionic neuron (third order neuron) Horner's syndrom

   a. Internal carotid artery → cavernous sinus → nasociliary nerve → long ciliary nerves → iris dilator

skull fracture
vasculopathic

More critical and most important cause for treatment for postganglionic Horner's syndrom is ICA DISSECTION (can cause a cloth)
If they headeach or any pain should go to energency hours away from stroke



H. Pharmacologic testing Horner's syndrom: develop super sensitivity

1. Apraclonidine 0.5 or 1% → to normal people nothing with Horner's sy. would dilated the pupil.
   a. Alpha adrenergic agonist

Also ptosis will go "away"
Muller muscles want to be stimulated as well

reversal of anisocoria
improvement of ptosis

VI. Physiologic anisocoria Normal anisocoria; at 20% of population

~ tend to fluctuate (stress related)
~ no dilation lag
~ < 1mm (subtle)

Anisocoria is a sign for Horner's syndrom.
But doesn't say you <u>have</u> to have it.